Week 7: Prompt Engineering

11
May'25

Generative AI for Office Productivity
Mentored Learning session
Topic: Prompt Engineering
JHU-Applied Gen AI-March 2025-A
Rishika.d@mygreatlearning.com

09:00AM - 11:00AM
Join session
Create/View Polls
Group of 28 students

https://olympus.mygreatlearning.com/courses/128577

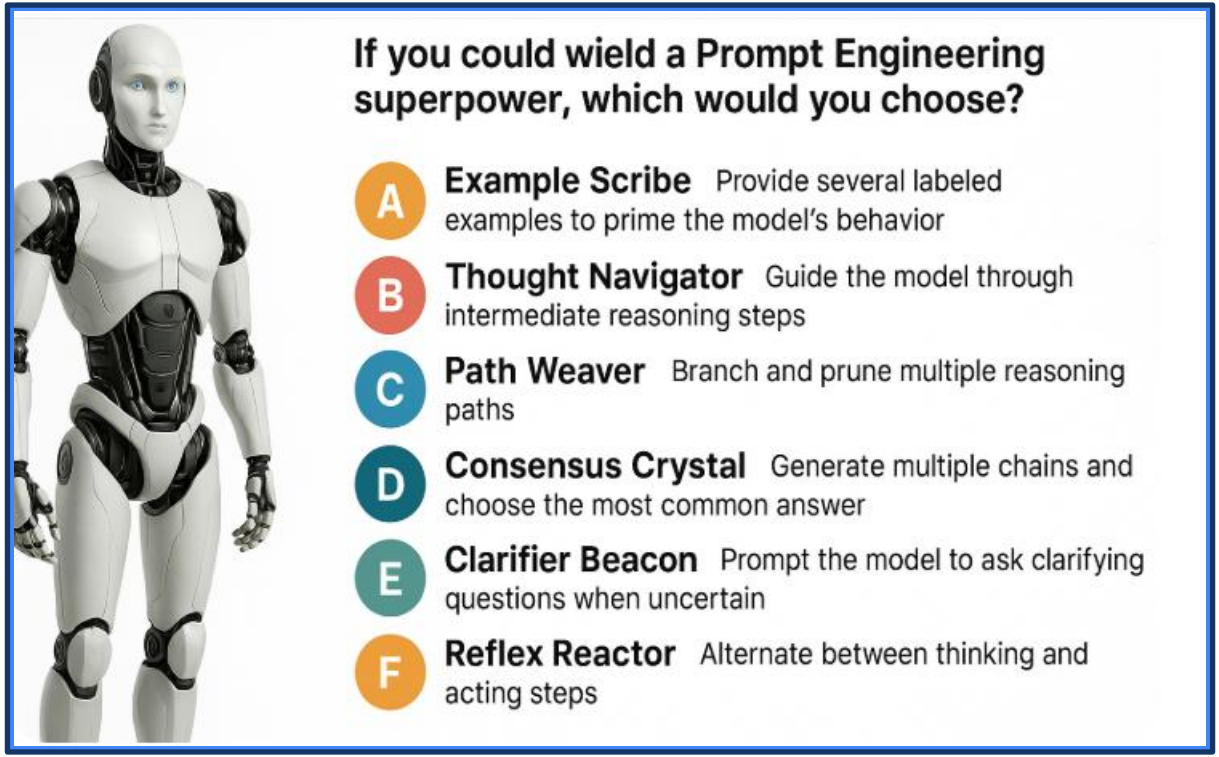
Mike Lively

Founder QuantumAI
Trainer, IT Evangelist,
Developer
@mikelively-quantumai

About Me

- Father of Nine
- Working on a PhD in GenAI
- Teaching AI for Johns Hopkins & GK
- Avid Hacker
- Flautist
- YMCA Gym Rat

ICE Breaker



Agenda

Prompt Engineering
- Intro
- Review

Case Study (Differential Diagnostic)

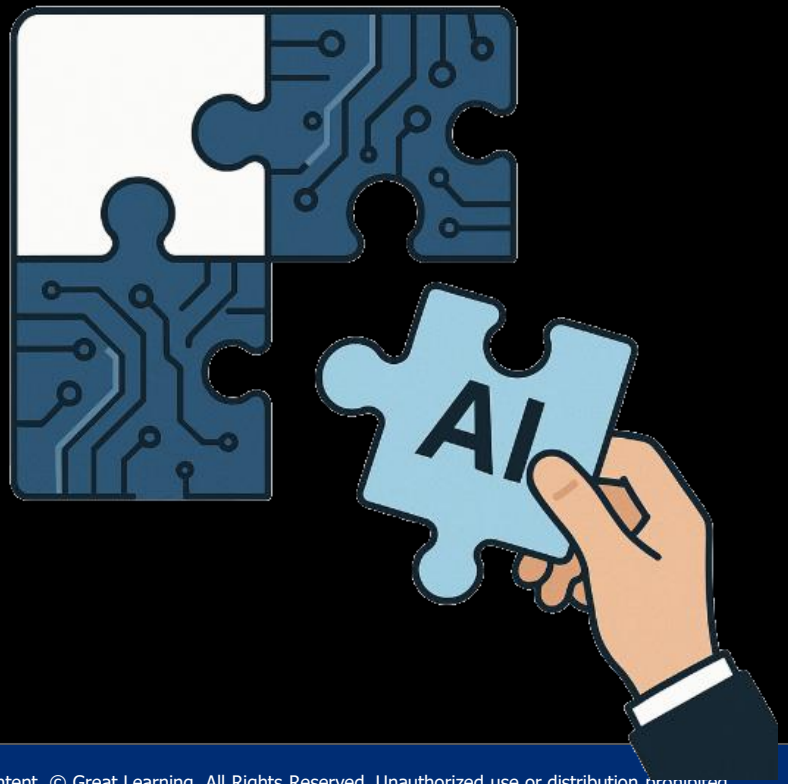
Walkthrough
- Data First
- Requirements
- Workflow
- Prompt Types
- Visualization (Updates)

DSL

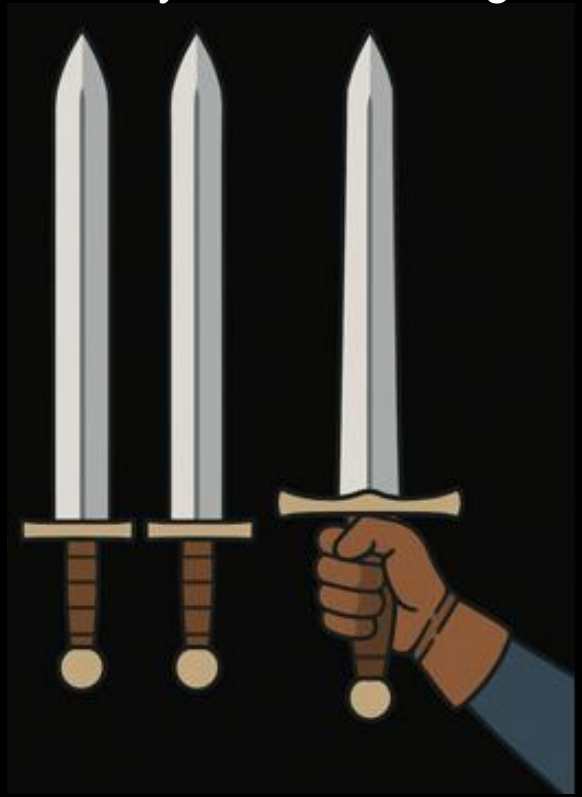
Conclusion & Insights

Session Summary

You're not making the puzzle pieces — you're putting them together!



You're not making swords — you’re mastering swordsmanship! m



Writing effective prompts

Best practices

1. **Be specific** about what you want Copilot/ChatGPT to do. Clear goals lead to better responses.
2. **Add some context** to help Copilot/ChatGPT understand what you're asking. Context makes the response more relevant.
3. **Provide some examples** for Copilot/ChatGPT to use. This helps ground the response in the right context.
4. Let Copilot/ChatGPT know how you want the response **to be formatted.** This sets clear expectations.

The art of the prompt

Goal	What do you want from Copilot?
Context	Why do you need it and who is involved?
Source(s)	What information or samples do you want Copilot to use?
Expectations	How should Copilot respond to best fulfill your request?

What's Next Thinking

What is the Prompt?

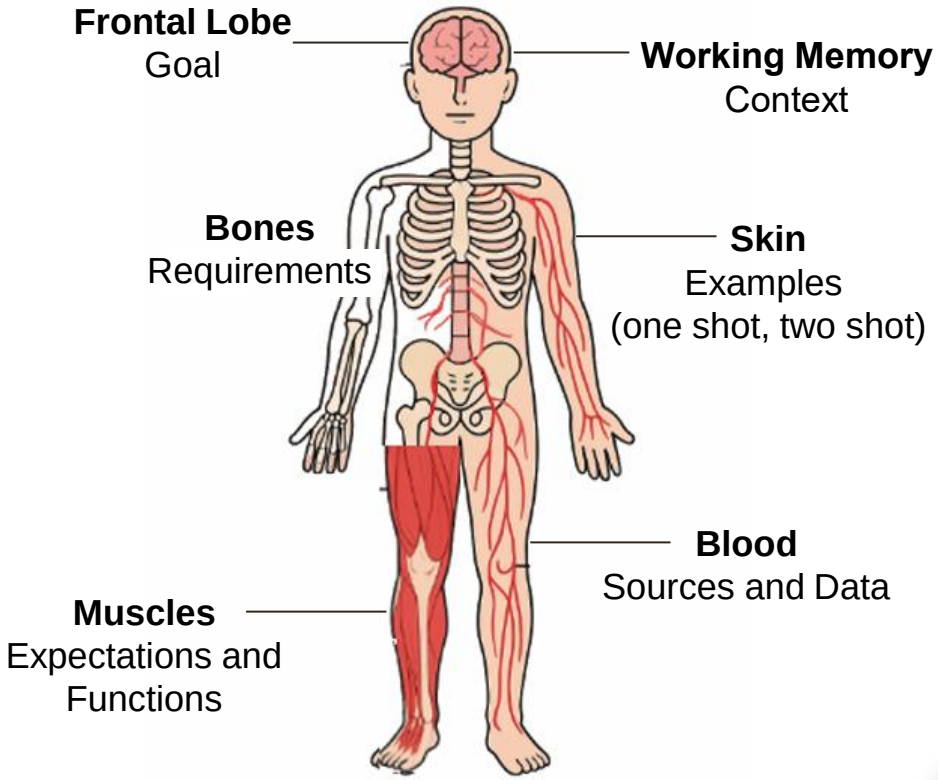
Practicing Progressive Prompts

Task	Option A (OK)	(Better)	Option B (Even Better)
Brainstorm newsletter topics	I need ideas for our company newsletter. Can you suggest some topics?	I'm working on our company newsletter. Can you suggest topics related to Efficiency tips?	I'm working on our company newsletter. Can you suggest topics related to efficiency tips for data analysts? Topics should include a concise headline and a short summary that highlights its relevance to data science.

Prompt Engineering

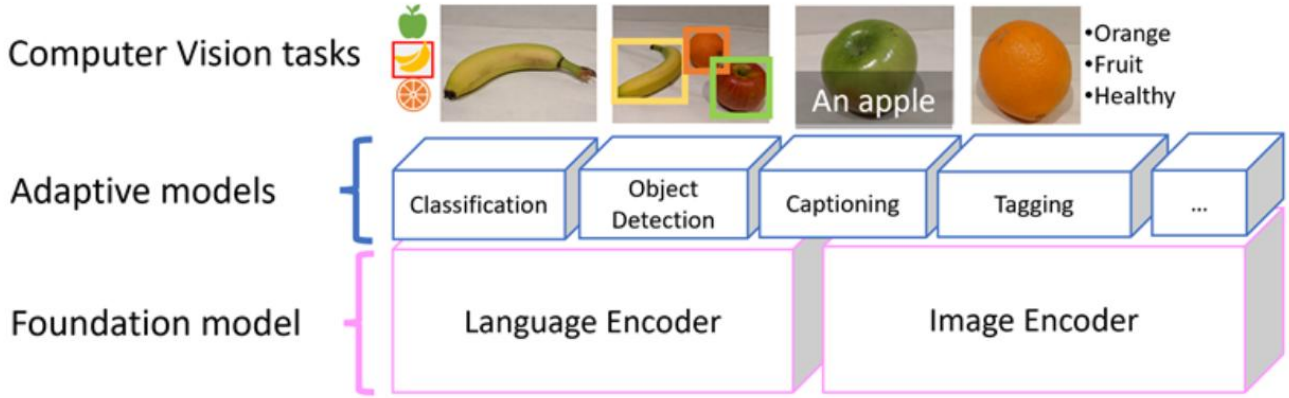
Think of a prompt as the human body

Goal
Context
Requirements
Expectations & Functions
Data & Sources
Examples



The Big Surprise

Multi-modal models (Surprise)



- A newer approach to modeling involves combining language and vision models that encode image and text data
- The model encapsulates semantic relationships between features extracted from the images and text extracted from related captions.
- A multi-modal model can be used as a *foundation* model for more specialized *adaptive* models.

Use Case

A local hardware store faces significant challenges in managing the billing process for customers. Currently, the store uses a manual checkout system, which is time-consuming, prone to human errors, and inefficient. This often results in customer dissatisfaction and potential revenue loss due to mismanagement or inaccurate billing.

To address this, the hardware store is exploring the implementation of a basic Cart Management System as a pilot solution to improve the purchasing experience. The system does not need to be overly complex but should provide essential functionalities to streamline operations and enhance customer satisfaction.

Example of a multi-modal prompt

Goal and Context: The primary goal is to create a program that digitally streamlines the billing process for a local hardware store that currently relies on manual methods, increasing efficiency, enhancing customer satisfaction, reducing operational errors, and laying the foundation for future growth.

Data

```
Inventory = [
    (101, "Hammer", 120),
    (102, "Screwdriver Set", 350),
    (103, "Drill Machine", 2500),
    (104, "Pliers", 180),
    (105, "Wrench", 220),
]

My_Cart = []
```

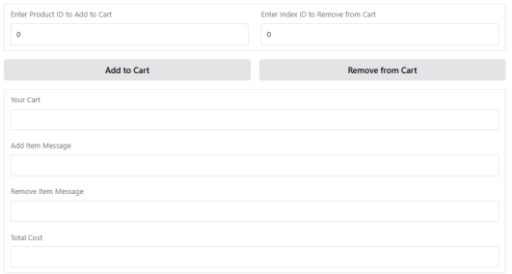
Requirements:

Add Items to the Cart: The store clerk should be able to select an item from the inventory and add it to the cart. The item's name and price should be pulled automatically based on the selected ID.
Remove Items from the Cart: The clerk should have the flexibility to remove individual tools from the cart before checkout to accommodate customer preferences or mistakes.
Calculate Total Bill: The system should calculate the total cost of items in the cart and display the final amount clearly to the customer for quick payment.

Expectations and Functions

The Gradio interface would include: A dropdown to select tools from inventory
An "Add Tool" buttonA second dropdown to select tools from the current cart
A "Remove Tool" buttonA display box showing the cart contentsA "Calculate Total" button with a result output

Interface Example



Wireframe (or image) to Webpage

Prompt 1: create an html input page from the image
Prompt 2: separate the html and the css
Prompt 3: change the styles.css file to make the webpage more appealing

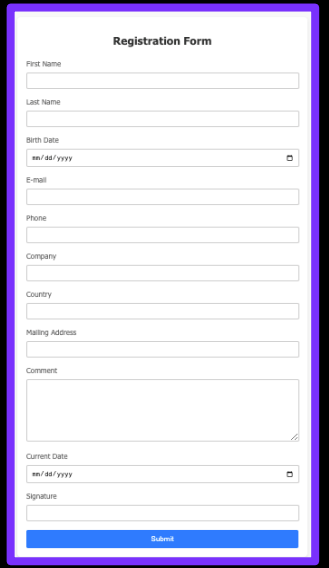
Wireframe (or Image)

HTML (54 lines)

+

CSS (52 lines)

Web Page



https://huggingface.co/spaces/eaglelandsonce/form

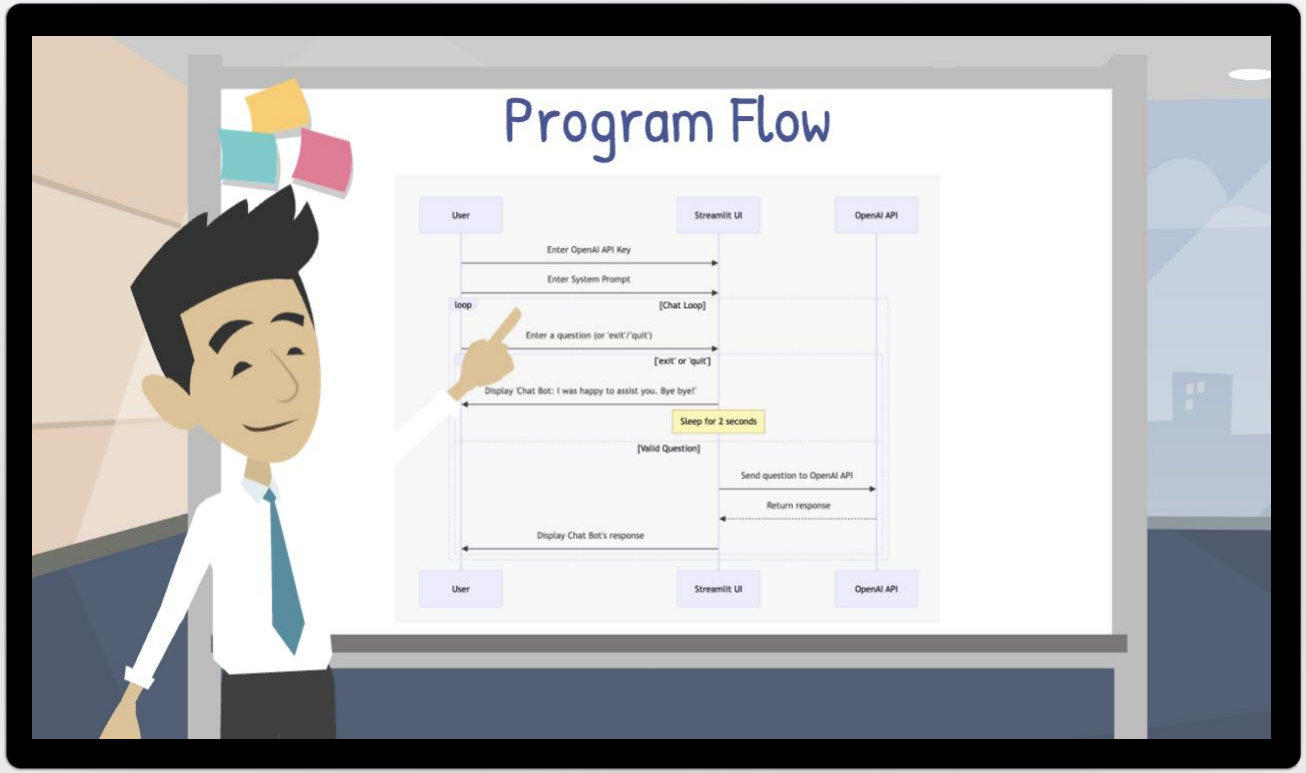
Wireframe (or image) to Webpage

Average coder writes about 10 to 100 lines of functional code a day.
You did this in 5 minutes: 9600% efficiency!

https://huggingface.co/spaces/eaglelandsonce/fern

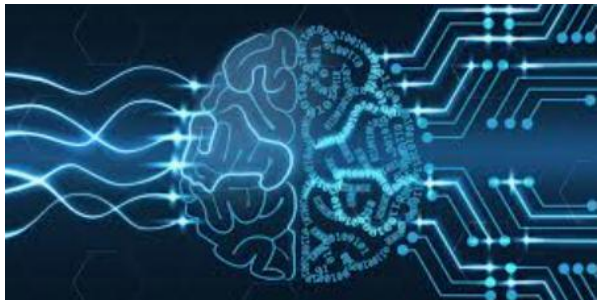
Scrum Master Superpower

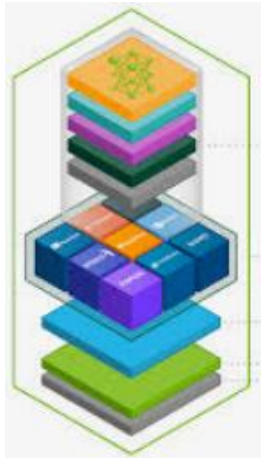
Image to Text (Including Code)



LLMs Captured Human Reasoning

Data Has Human Reasoning

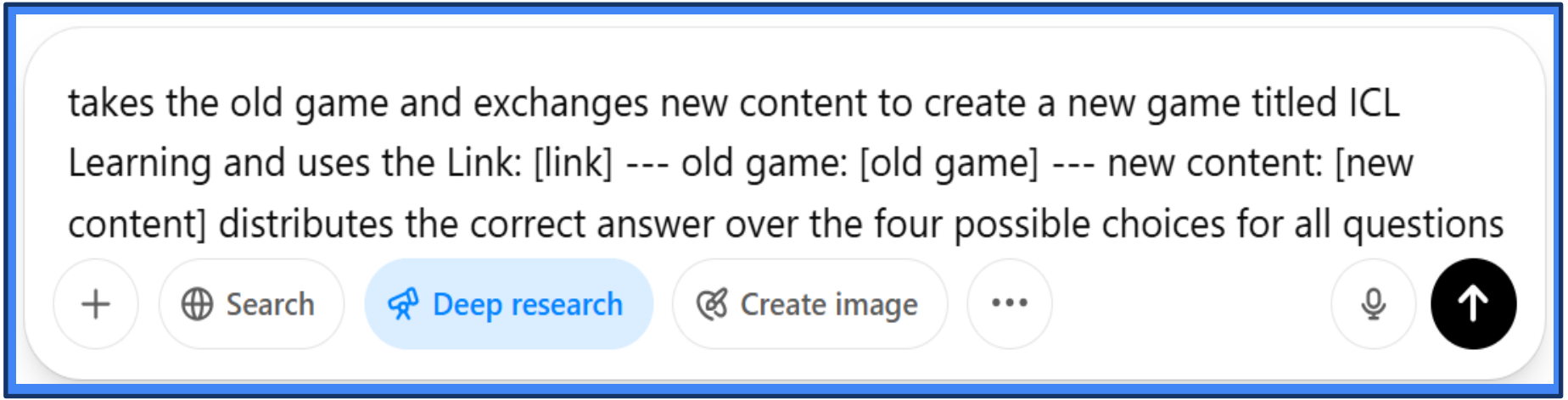




Trained

LLMs captured not just data, but also human reasoning!

Power of ICL ("In-Context Learning")



https://www.linkedin.com/pulse/incontext-learning-guided-expedition-contextual-ai-michael-lively-yas9e/

https://huggingface.co/spaces/eaglelandsonce/ICL

Power of ICL ("In-Context Learning")

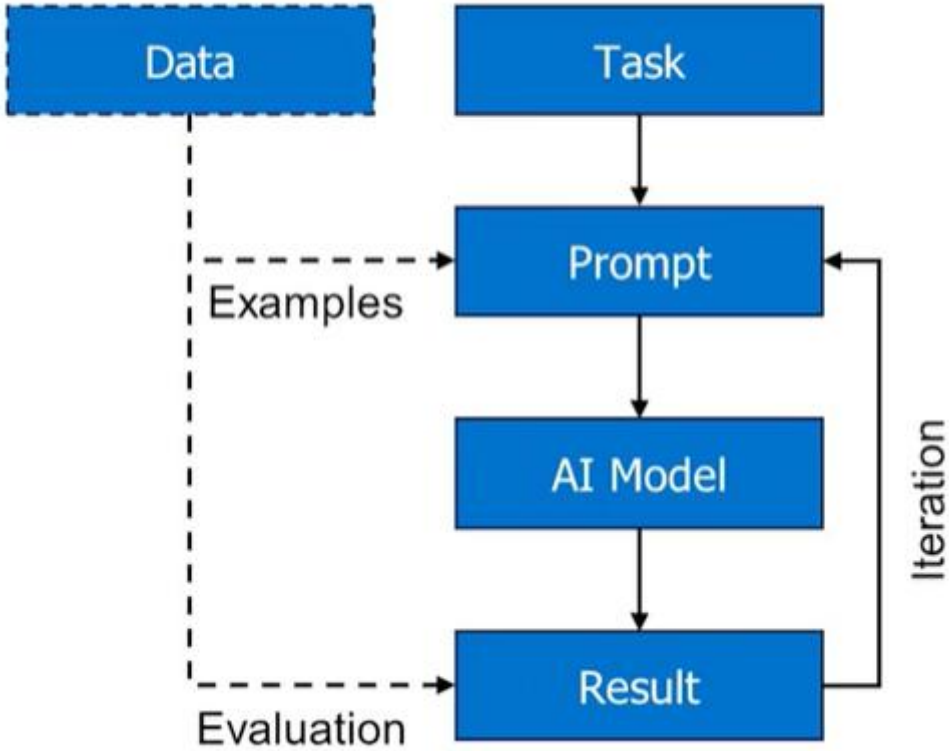
In-Context

https://huggingface.co/spaces/eaglelandsonce/NLP_Millionaire

https://www.linkedin.com/pulse/incontext-learning-guided-expedition-contextual-ai-michael-lively-yas9e/

Biggest issue: Over reliance on AI!

In this session:



Prompting Patters

Match the Prompting Pattern

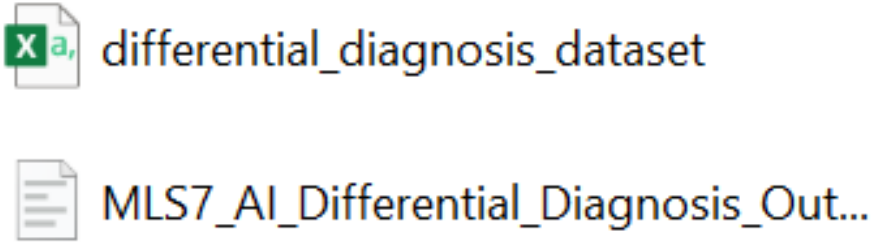
Patterns	Definitions
Few-Shot Prompting	Alternate between "thinking" and "acting" (e.g., API calls) with observations guiding next steps.
Chain-of-Thought Prompting	Guide the model through step-by-step reasoning before producing a final answer.
Tree-of-Thought Prompting	Provide a handful of carefully chosen examples to "prime" the model's behavior.
Self-Consistency Prompting	Have the model ask clarifying questions when uncertain, incorporating feedback.
Active Prompting	Branch into multiple reasoning paths, then prune or merge them to find the best solution.
ReAct Prompting	Generate multiple independent reasoning chains and pick the most common answer.

Hint

Let the model ask you a question if it's not sure.

https://huggingface.co/spaces/eaglelandsonce/Prompt_Matching

Files for Today



Three Options: Google Colab, VSCode local, or GitHub Codespaces

https://huggingface.co/spaces/eaglelandsonce/Batch_Sentiment_Analysis

Diagnosis Case Study

In the healthcare domain, timely and accurate diagnosis of medical conditions is crucial for effective treatment and improved patient outcomes. However, the differential diagnosis process is often complex and time-consuming, especially when multiple symptoms overlap across different medical conditions.

Medical practitioners face the challenge of manually analyzing patient symptoms, which can lead to delays, inefficiencies, and potential misdiagnoses. Traditional methods of diagnosis rely heavily on the experience of the practitioner, making the process prone to variability and subjectivity.

To address these challenges, the healthcare sector is increasingly exploring AI-based solutions to improve diagnostic processes. Leveraging Generative AI models can significantly enhance the accuracy, speed, and consistency of differential diagnosis by understanding the relationship between patient symptoms and possible medical conditions.

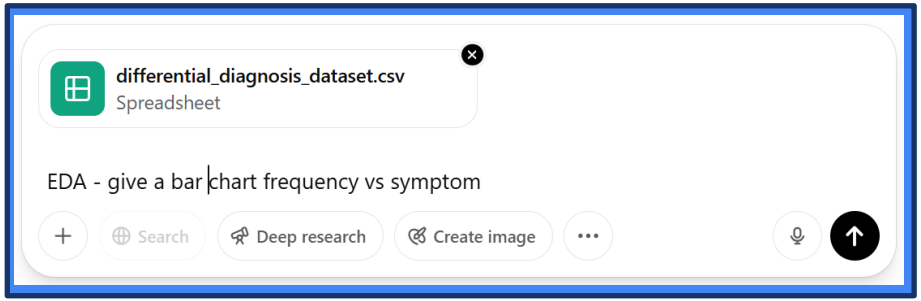
Key Business Outcomes

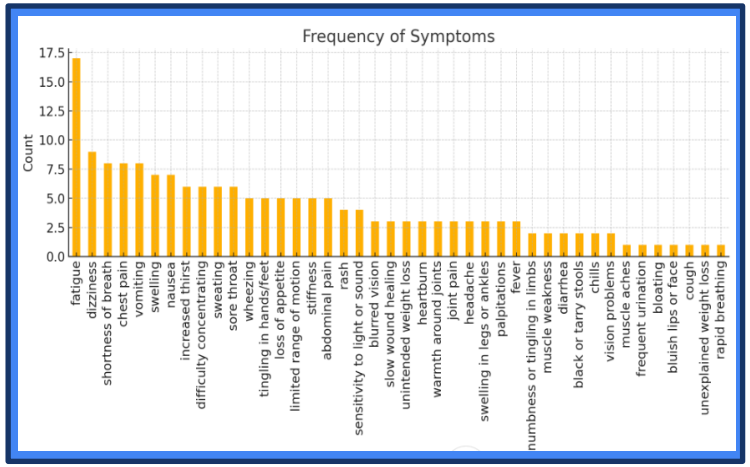
The goal of this case study is to build an AI-powered differential diagnosis system that serves as a decision support tool for healthcare professionals, ultimately improving diagnostic accuracy and patient care.

Key Business Outcomes

✅ **Improved Diagnostic Accuracy:** Generate differential diagnosis suggestions with higher accuracy based on symptom patterns.
✅ **Faster Decision-Making:** Reduce the time taken to analyze symptoms and suggest probable conditions.
✅ **Enhanced Patient Care:** Enable healthcare professionals to provide faster and more effective treatment.
✅ **Consistency in Diagnosis:** Standardize the diagnostic process, reducing variability in medical decisions.
✅ **Demonstrate Prompting Techniques:** Illustrate different prompting methods like Few-Shot, Chain of Thought (CoT), and Tree of Thought to teach AI-based decision-making.

Data First EDA





Objective

The primary objective is to develop an **AI-powered differential diagnosis system** that:

- Understands the **relationship** between patient symptoms and possible medical conditions.
- Utilizes **LLaMA** models for generating **diagnosis suggestions**.
- Applies **Few-Shot Prompting**, **Chain of Thought (CoT)**, and **Tree of Thought (ToT)** techniques for symptom-based predictions.
- Delivers **consistent and reliable** differential diagnosis outcomes.

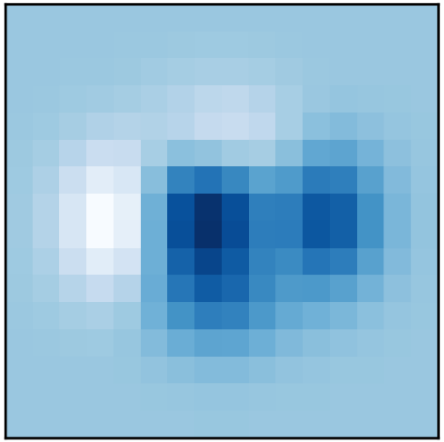
Installs

- **numpy, scipy:** For numerical computations.
- **huggingface_hub:** To interact with the Hugging Face model hub.
- **datasets:** For loading benchmark datasets.
- **evaluate:** For performance evaluation.
- **matplotlib, seaborn:** For data visualization.

Matplotlib

matplotlib

Plot types | User guide | Tutorials | Examples | Reference | Contribute | Releases



imshow(Z)

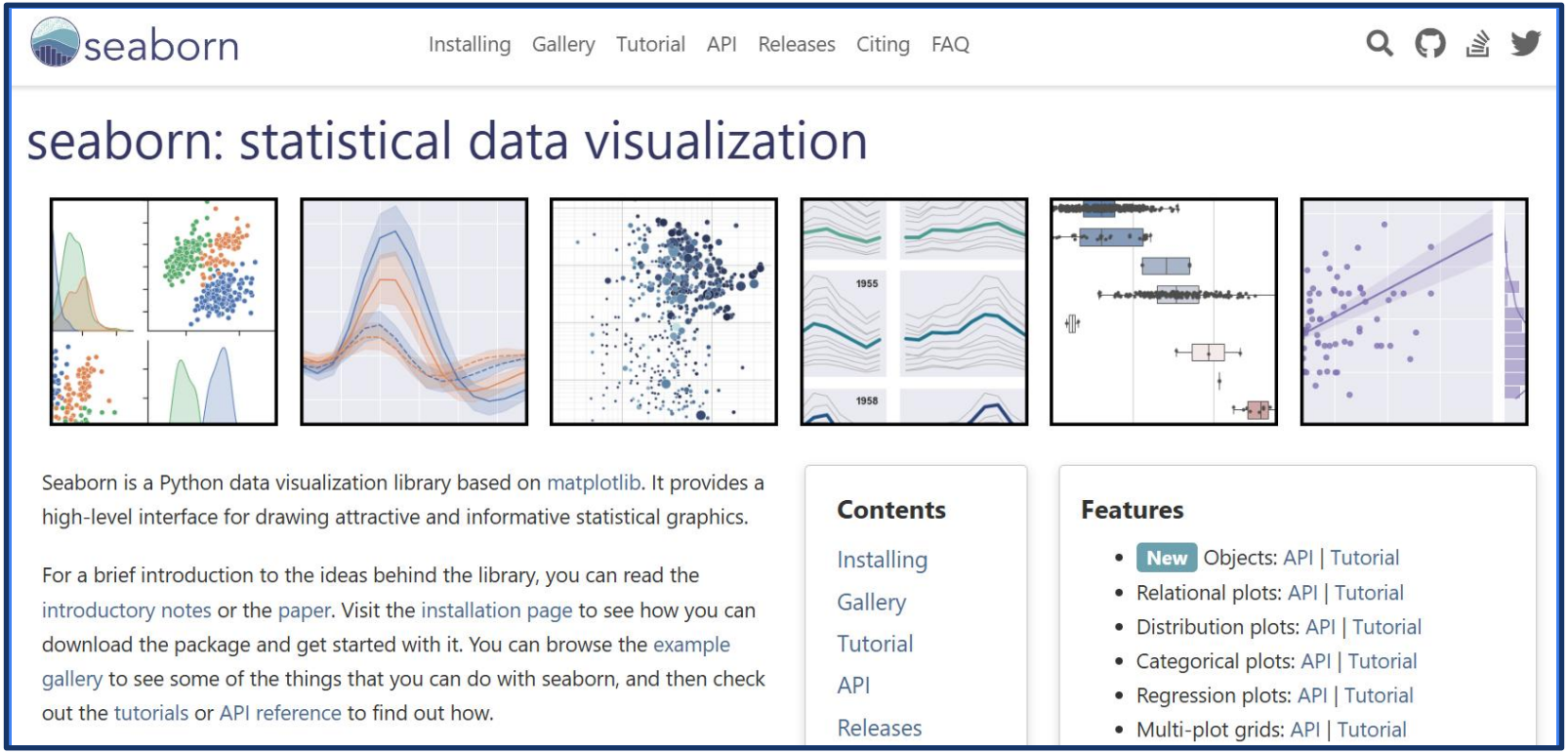
Matplotlib: Visualization with Python

Matplotlib is a comprehensive library for creating static, animated, and interactive visualizations in Python. Matplotlib makes easy things easy and hard things possible.

- Create publication quality plots.
- Make interactive figures that can zoom, pan, update.
- Customize visual style and layout.
- Export to many file formats.
- Embed in JupyterLab and Graphical User Interfaces.
- Use a rich array of third-party packages built on Matplotlib.

https://matplotlib.org/

Seaborn



https://seaborn.pydata.org/

Huggingface_hub

Project description

Release history

Download files

Verified details

These details have been verified by PyPI

Maintainers


celinah


chaumond


lysandre


Wauplin

huggingface_hub

The official Python client for the Huggingface Hub.

English | Deutsch | हिंदी | 한국어 | 中文（简体）

Documentation: https://hf.co/docs/huggingface_hub

Source Code: https://github.com/huggingface/huggingface_hub

Welcome to the huggingface_hub library

The `huggingface_hub` library allows you to interact with the Hugging Face Hub, a platform democratizing open-source Machine Learning for creators and collaborators. Discover pre-trained

https://pypi.org/project/huggingface-hub/

Solution Approach

1. Environment Setup & Installation

- Install necessary libraries and **verify GPU** availability using `nvidia-smi`.
- Set up Colab notebook environment for running the **differential diagnosis** pipeline.

2. Model Loading & Configuration

- Install `llama-cpp-python` with GPU support.
- Load **Meta-Llama-3-8B-Instruct-GGUF** model using `llama-cpp-python` for inference.
- Configure model parameters such as **temperature**, **max tokens**, and **top-k sampling** for generating responses.

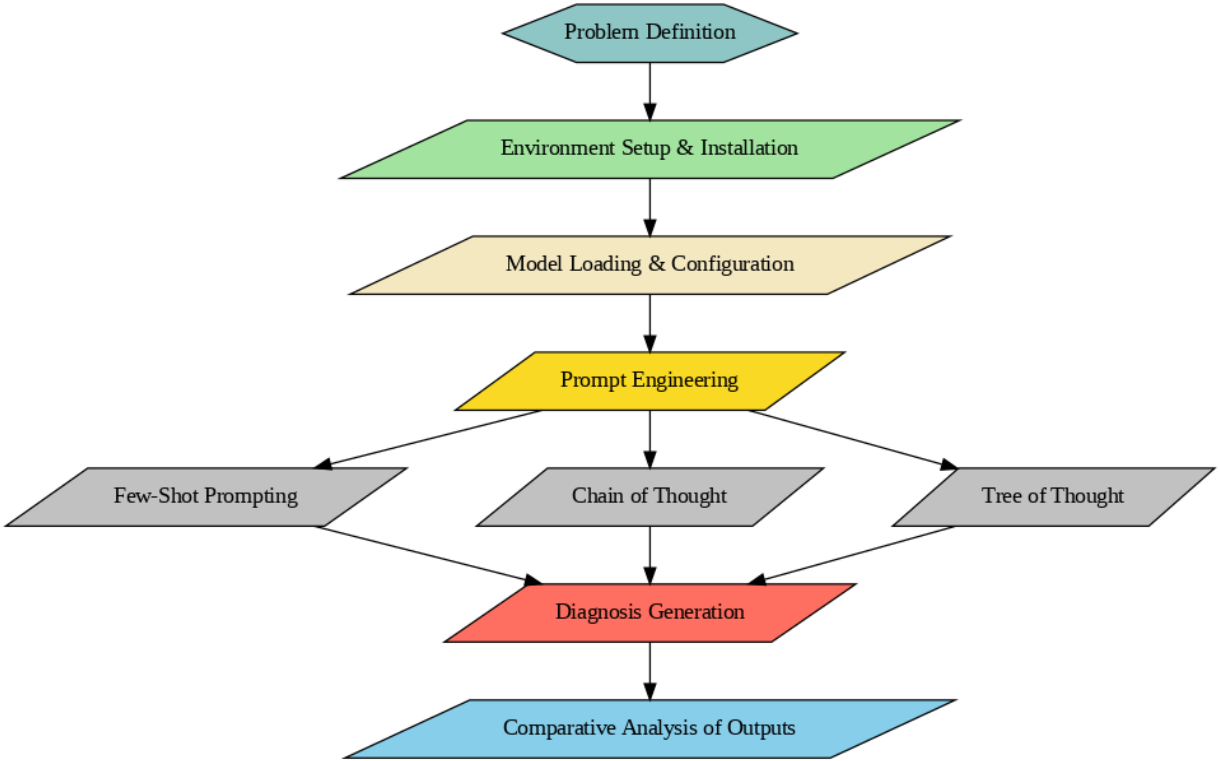
3. Prompt Engineering

- **Few-Shot Prompting:** Include 2-3 examples of symptom-diagnosis pairs to guide the model.
- **Chain of Thought (CoT):** Formulate prompts that instruct the model to explain its reasoning step-by-step before arriving at a diagnosis.
- **Tree of Thought (ToT):** Design prompts that guide the model through branching decision points based on individual symptoms.

4. Inference & Output Generation

- Generate **differential diagnosis suggestions** based on patient symptoms.
- Format outputs with **numbered diagnostic possibilities** and reasoning.
- **Compare model** responses across Few-Shot, CoT, and ToT techniques

Solution Workflow



Few-Shot

```
SYSTEM:
You are an AI medical assistant specializing in differential diagnosis.
Generate the most likely list of diagnoses based on examples.

USER: A 45-year-old male, fever, cough, fatigue.
SYSTEM: [Flu, COVID-19, Pneumonia]

USER: A 30-year-old female, severe abdominal pain, nausea.
SYSTEM: [Appendicitis, Gallstones, Gastritis]

USER: A 10-year-old female, wheezing.
SYSTEM: [Asthma, Respiratory Infection]

USER: + "A 35-year-old male, fever, wheezing, nausea."
```

Chain of Thought (CoT)

```
SYSTEM:
You are a medical expert performing differential diagnosis through step-by-step reasoning.
1. Analyze the symptoms and list underlying possible conditions.
2. Explain each possibility step-by-step.
3. Provide the final list of probable diagnoses.

Provide the final output in the format provided below.
OUTPUT:
  Explanation of reasoning:
  "
  - Fatigue is a common symptom in many conditions.
  - For a 50-year-old male, fatigue can indicate cardiovascular or metabolic issues.
  - Common diagnoses for fatigue:
    - Anemia
    - Diabetes
    - Thyroid disorder
    - Heart disease "

  Most probable diagnosis: [Diagnosis 1, Diagnosis 2, Diagnosis 3]

USER: + "A 35-year-old male, fever, wheezing, nausea."
```

Tree of Thought (ToT)

```
 SYSTEM: Simulate a conversation among three expert doctors discussing
differential diagnosis.

    Each doctor will present their reasoning step-by-step for
differential diagnosis based on the symptoms provided.
    The experts will cross-verify each other's suggestions and finally
agree on the top 3 most probable diagnoses.

    Provide the final output in the format provided below.
    OUTPUT:
      Most probable diagnosis: [Diagnosis 1, Diagnosis 2, Diagnosis 3]


    USER: + "A 35-year-old male, fever, wheezing, nausea."
```

DSL


```
SYSTEM: Simulate a conversation among three expert doctors discussing
differential diagnosis.

    Each doctor will present their reasoning step-by-step for
differential diagnosis based on the symptoms provided.
    The experts will cross-verify each other's suggestions and finally
agree on the top 3 most probable diagnoses.

    Provide the final output in the format provided below.
    OUTPUT:
      Most probable diagnosis: [Diagnosis 1, Diagnosis 2, Diagnosis 3]


    USER: + "A 35-year-old male, fever, wheezing, nausea."
```

SYS:3Dr(diff-dx):r→v→c;OUT:[dx1,dx2,dx3] USR:35M,fvr,whzzng,ns

DSL

SYS:3Dr(diff-dx):r→v→c;OUT:[dx1,dx2,dx3] USR:35M,fvr,whzzng,ns

1. `SYS:`

   Indicates this is a **system-level** instruction—i.e. setting up ChatGPT's behavior rather than user dialogue.

2. `3Dr(diff-dx)`

   - `3Dr` → "Simulate **3 Doctors**"
   - `(diff-dx)` → Each doctor performs a **differential diagnosis**

DSL

SYS:3Dr(diff-dx):r→v→c;OUT:[dx1,dx2,dx3] USR:35M,fvr,whzzng,ns

3. `:r→v→c`

   Defines the **three reasoning phases**, in order:

   1. **r** (Reason) — each doctor presents their initial reasoning
   2. **v** (Verify) — they critique and verify each other's suggestions
   3. **c** (Consense) — they reach consensus on the top diagnoses

4. `;OUT:[dx1,dx2,dx3]`

   Specifies the **output format**: a list of three diagnoses in the array `[...]`, labeled `dx1`, `dx2`, `dx3`.

DSL

SYS:3Dr(diff-dx):r→v→c;OUT:[dx1,dx2,dx3] USR:35M,fvr,whzzng,ns

5. `USR:35M,fvr,whzzng,ns`

   Feeds in the **user's patient data** as a CSV of shorthand fields:

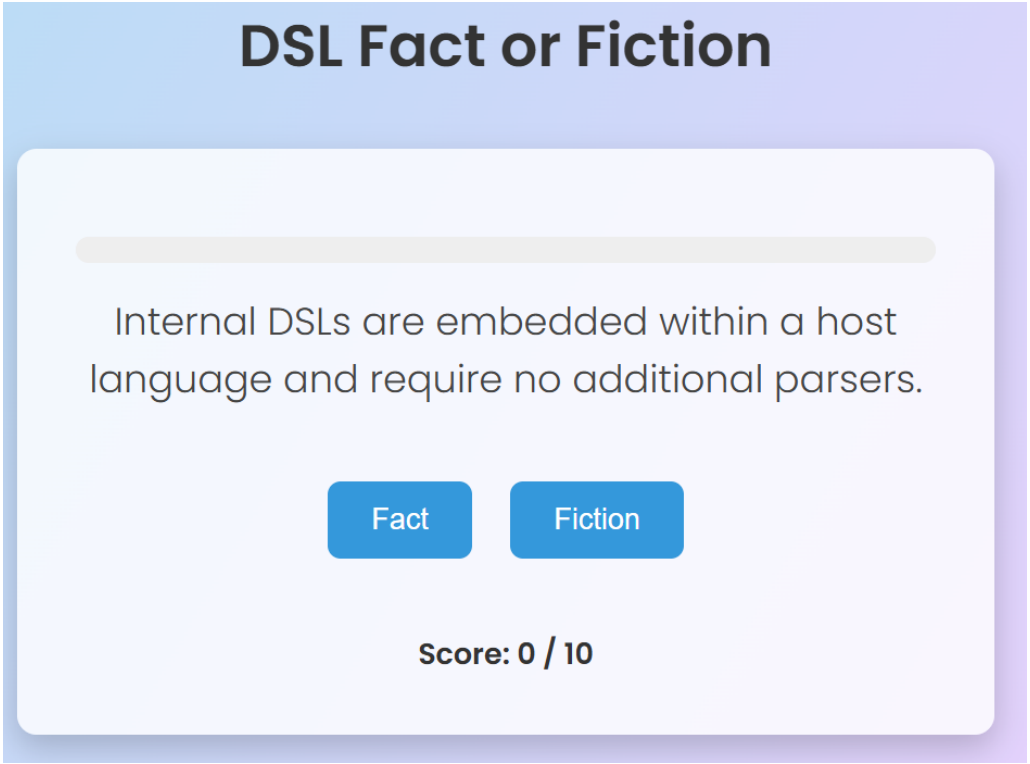
   - `35M` → 35-year-old male
   - `fvr` → fever
   - `whzzng` → wheezing
   - `ns` → nausea

DSL

SYS:3Dr(diff-dx):r→v→c;OUT:[dx1,dx2,dx3] USR:35M,fvr,whzzng,ns

As a system prompt, simulate a panel of three doctors doing a differential diagnosis in three steps—Reason → Verify → Consensus—then return their final top three diagnoses for a 35-year-old man with fever, wheezing, and nausea."

Will AI start speaking and programming in its own language – one we can not understand?



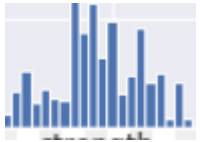
https://huggingface.co/spaces/eaglelandsonce/DSL_Fact_or_Fiction

https://www.linkedin.com/pulse/domain-specific-languages-dsls-prompt-engineering-michael-lively-kuk3e

DSL

Why is DSL a more powerful prompt?

DSL is like conceptual featurization it will only get better with time where NLP prompts will change with newer models!



Will AI Robots develop a language we can not understand?

Summary

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- Review

Case Study (Differential Diagnostic)
Walkthrough
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- Visualization (Updates)

DSL
Conclusion & Insights
Session Summary

JOHNS HOPKINS
WHITING SCHOOL
of ENGINEERING

AI-Conveyor Belt

Great Learning
POWER AHEAD

Thoughts? Questions?

Appendix

JOHNS HOPKINS
WHITING SCHOOL of ENGINEERING

Great Learning
POWER AHEAD

About Me

- Teaching for Microsoft, Global Knowledge & Great Learning.
- Prompt Engineering for Outliers and Snorkel.

- Military (USAF Captain), Government, Education & Industry

- PMI.ACP, PSM, LSSBB, CPCU, ACS
- 24 Cloud Certifications: Azure MCT, AZ-305, AZ-700, AZ-400, AZ-500, AZ-104, AZ-900, AI-102, AI-900, DP-203, DP-100, DP-900, DP-500, DP-600, PL-300, AWS. DOP, ANS, DAS, MLS, SCS, SAP Google CDL, ACE, PCA
- Databricks Fundamentals, Azure Databricks Platform Architect

- Avid Generative AI Hacker (winning 8 global Hackathons)
- Keynote Speaker: PMISWO Summit 2024, VP Professional Development PMI SWO Chapter

- BA Mathematics, BS Chemistry, BSEE in Electrical Engineering, MS in Physics, Ph.D. “ABD” Nuclear Physics, (pursuing) Ph.D. in Generative AI & ML, Donovan Scholar at University of Kentucky

- LinkedIn: https://www.linkedin.com/in/mikelively-quantumai/
- Multi-Cloud Meetup, Hugging Face & Lablab.ai